

ML-09: Validation and Research Claim Audit

NOTEBOOK: [work/notebooks/w06_validation_audit.ipynb](#) | AUTHOR: Abdul Sami | TRACK: Machine Learning

1. Paper Findings & Constructive Methodology Questions

- **Finding 1 (CTR Opportunity Signal Correlation):** *Where does the 'opportunity' label originate, and how are confounding intent shifts controlled?* Questioned whether zero-click SERP features (Featured Snippets / Knowledge Panels) confound CTR underperformance vs metadata quality.
- **Finding 2 (Cross-Domain Generalizability):** *Does random train/test validation split leak domain-specific artifacts?* Emphasized the necessity of domain-grouped validation (GroupKFold) to test true out-of-domain generalizability.

2. My Model Under an Honest Split (Before vs After)

Split Design	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Before (Random Holdout Split)	96.67%	0.917	0.688	0.786	0.995
After (Domain-Grouped Split)	95.83%	0.818	0.750	0.783	0.983

3. Feature Leakage Audit & Failure Case Inspection

- **Temporal Audit:** Verified all 5 features (past_impressions_30d, historical_avg_position, historical_ctr, content_word_count, days_since_last_update) are knowable at decision moment ($t \leq t_0$).
- **Error Profile:** False positives occur primarily on rank 7–9 border pages on high-authority domains due to SERP ad layout placement rather than metadata quality.

4. Claim Rewrites (Safe Claim Language)

- **Inflated Claim (Before):** 'Our ML model guarantees 95%+ accuracy in identifying pages that will double traffic.'
- **Public-Safe Claim (After):** 'In offline evaluation on a 30% holdout test set, the Gradient Boosting model **observed** an F1-Score of 0.783 and an ROC-AUC of 0.983 under domain-stratified splits.'
- **Inflated Claim (Before):** 'Rewriting meta descriptions automatically restores lost CTR.'
- **Public-Safe Claim (After):** 'The model serves as a **directional decision-support signal** to prioritize URLs where historical CTR lags position benchmarks.'